Tutorial 7

Feature Extraction vs. Fine-Tuning on CIFAR-10 using PyTorch

Task 1: Feature Extraction with VGG16

Setup

- Model: VGG16 pretrained on ImageNet with all backbone layers frozen.
- Only the new classifier head was trained on 40,970 trainable parameters.
- Epochs: 10 , Dataset: CIFAR-10 , 5,000 training / 1,000 test samples

Training Log

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Training samples : 5000
Test samples     : 1000
============================================================
STEP 3A: Feature Extraction — VGG16
============================================================
Trainable parameters (VGG16 Feature Extraction): 40970
[VGG16-FE] Epoch 01/10 | Train Loss: 1.0434 Acc: 0.6338 | Val Loss: 0.6992 Acc: 0.7460
[VGG16-FE] Epoch 02/10 | Train Loss: 0.7937 Acc: 0.7180 | Val Loss: 0.7359 Acc: 0.7400
[VGG16-FE] Epoch 03/10 | Train Loss: 0.7941 Acc: 0.7308 | Val Loss: 0.7822 Acc: 0.7190
[VGG16-FE] Epoch 04/10 | Train Loss: 0.8227 Acc: 0.7168 | Val Loss: 0.6838 Acc: 0.7510
[VGG16-FE] Epoch 05/10 | Train Loss: 0.8092 Acc: 0.7288 | Val Loss: 0.6966 Acc: 0.7470
[VGG16-FE] Epoch 06/10 | Train Loss: 0.7900 Acc: 0.7290 | Val Loss: 0.6938 Acc: 0.7460
[VGG16-FE] Epoch 07/10 | Train Loss: 0.8469 Acc: 0.7258 | Val Loss: 0.7037 Acc: 0.7550
[VGG16-FE] Epoch 08/10 | Train Loss: 0.8402 Acc: 0.7334 | Val Loss: 0.7061 Acc: 0.7610
[VGG16-FE] Epoch 09/10 | Train Loss: 0.7660 Acc: 0.7466 | Val Loss: 0.7398 Acc: 0.7430
[VGG16-FE] Epoch 10/10 | Train Loss: 0.7956 Acc: 0.7450 | Val Loss: 0.7156 Acc: 0.7470

VGG16-FE — Best Val Acc: 0.7610 | Time: 7.7 min
```

Results and Analysis

- Best Validation Accuracy: 76.10% and Test Accuracy: 76.10%
- Training loss showed minimal improvement after Epoch 2, remaining in the range of 0.77 to 1.04 throughout.
- Validation accuracy peaked at 76.10% in Epoch 8 and did not improve further.
- With all backbone layers frozen, the classifier head had limited capacity to adapt the fixed ImageNet features to CIFAR-10.
- Training completed in 7.7 minutes due to the very low number of trainable parameters.

Since the backbone was entirely frozen, the model relied on ImageNet feature representations without any adaptation. The pretrained features transferred reasonably well, producing a 76.10% baseline. However, the stagnant training loss beyond Epoch 2 indicates the classifier head alone cannot bridge the gap between ImageNet and CIFAR-10 characteristics.

Task 2: Fine-Tuning with ResNet50

Setup

- Model: ResNet50 pretrained on ImageNet with the last 5 layers unfrozen for fine-tuning.
- Trainable parameters: 23,303,178.
- Epochs: 15 and Same CIFAR-10 split.

Training Log

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============================================================
STEP 3B: Fine-Tuning — ResNet50 (Last 5 Layers)
============================================================
Trainable parameters (ResNet50 Fine-Tuning): 23303178
[ResNet50-FT] Epoch 01/15 | Train Loss: 1.6157 Acc: 0.4782 | Val Loss: 0.8799 Acc: 0.7710
[ResNet50-FT] Epoch 02/15 | Train Loss: 0.7568 Acc: 0.7768 | Val Loss: 0.5117 Acc: 0.8530
[ResNet50-FT] Epoch 03/15 | Train Loss: 0.5445 Acc: 0.8290 | Val Loss: 0.3885 Acc: 0.8790
[ResNet50-FT] Epoch 04/15 | Train Loss: 0.4097 Acc: 0.8678 | Val Loss: 0.3420 Acc: 0.8890
[ResNet50-FT] Epoch 05/15 | Train Loss: 0.3475 Acc: 0.8888 | Val Loss: 0.3048 Acc: 0.8950
[ResNet50-FT] Epoch 06/15 | Train Loss: 0.3033 Acc: 0.8992 | Val Loss: 0.3024 Acc: 0.8980
[ResNet50-FT] Epoch 07/15 | Train Loss: 0.2745 Acc: 0.9092 | Val Loss: 0.2693 Acc: 0.9040
[ResNet50-FT] Epoch 08/15 | Train Loss: 0.2560 Acc: 0.9192 | Val Loss: 0.2479 Acc: 0.9140
[ResNet50-FT] Epoch 09/15 | Train Loss: 0.2197 Acc: 0.9288 | Val Loss: 0.2495 Acc: 0.9180
[ResNet50-FT] Epoch 10/15 | Train Loss: 0.2107 Acc: 0.9308 | Val Loss: 0.2448 Acc: 0.9170
[ResNet50-FT] Epoch 11/15 | Train Loss: 0.1794 Acc: 0.9402 | Val Loss: 0.2179 Acc: 0.9260
[ResNet50-FT] Epoch 12/15 | Train Loss: 0.1827 Acc: 0.9384 | Val Loss: 0.2377 Acc: 0.9240
[ResNet50-FT] Epoch 13/15 | Train Loss: 0.1625 Acc: 0.9488 | Val Loss: 0.2323 Acc: 0.9240
[ResNet50-FT] Epoch 14/15 | Train Loss: 0.1359 Acc: 0.9546 | Val Loss: 0.2294 Acc: 0.9200
[ResNet50-FT] Epoch 15/15 | Train Loss: 0.1353 Acc: 0.9560 | Val Loss: 0.2211 Acc: 0.9320

ResNet50-FT — Best Val Acc: 0.9320 | Time: 14.9 min
```

Results and Analysis

- Best Validation Accuracy: 93.20% | Test Accuracy: 93.20%
- Training accuracy increased steadily from 47.82% at Epoch 1 to 95.60% at Epoch 15.
- Validation loss decreased consistently from 0.8799 to 0.2211 with no divergence observed.
- The difference between training and validation accuracy remained small throughout, indicating no overfitting.
- Training completed in 14.9 minutes.

Unfreezing the last 5 layers of ResNet50 allowed the model to adapt its deeper feature representations to CIFAR-10 image characteristics. Both training and validation metrics improved continuously across all 15 epochs, suggesting that additional epochs could yield further gains. ResNet50 Fine-Tuning produced the highest accuracy across all models evaluated.

Task 3 : Feature Extraction with ResNet50

Setup

- Model: ResNet50 pretrained on ImageNet with all backbone layers frozen.
- Only the classifier head was trained on 20,490 trainable parameters.
- Epochs: 10 and Early stopping applied (patience = 5).

Training Log

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============================================================
TASK: Feature Extraction — ResNet50
============================================================
Trainable parameters (ResNet50 Feature Extraction): 20490
[ResNet50-FE] Epoch 01/10 | Train Loss: 1.4350 Acc: 0.5340 | Val Loss: 0.9719 Acc: 0.6920
[ResNet50-FE] Epoch 02/10 | Train Loss: 0.9718 Acc: 0.6706 | Val Loss: 0.7495 Acc: 0.7690
[ResNet50-FE] Epoch 03/10 | Train Loss: 0.8712 Acc: 0.6980 | Val Loss: 0.7754 Acc: 0.7400
[ResNet50-FE] Epoch 04/10 | Train Loss: 0.8455 Acc: 0.7150 | Val Loss: 0.6913 Acc: 0.7890
[ResNet50-FE] Epoch 05/10 | Train Loss: 0.8346 Acc: 0.7160 | Val Loss: 0.6436 Acc: 0.7910
[ResNet50-FE] Epoch 06/10 | Train Loss: 0.8355 Acc: 0.7128 | Val Loss: 0.6517 Acc: 0.7620
[ResNet50-FE] Epoch 07/10 | Train Loss: 0.7965 Acc: 0.7222 | Val Loss: 0.6369 Acc: 0.7880
[ResNet50-FE] Epoch 08/10 | Train Loss: 0.8287 Acc: 0.7220 | Val Loss: 0.6414 Acc: 0.7740
[ResNet50-FE] Epoch 09/10 | Train Loss: 0.8131 Acc: 0.7202 | Val Loss: 0.6502 Acc: 0.7730
[ResNet50-FE] Epoch 10/10 | Train Loss: 0.7985 Acc: 0.7294 | Val Loss: 0.6865 Acc: 0.7640
  ↳ Early stopping triggered at epoch 10

ResNet50-FE — Best Val Acc: 0.7910 | Time: 6.0 min
```

Results and Analysis

- Best Validation Accuracy: 79.10% and Test Accuracy: 79.10%
- Validation accuracy peaked at 79.10% in Epoch 5 and became inconsistent beyond that point.
- Early stopping triggered at Epoch 10, restoring the best weights from Epoch 5.
- Fastest training run at 6.0 minutes.

Comparison of ResNet50-FE vs VGG16-FE vs Custom CNN

- ResNet50-FE (79.10%) outperformed VGG16-FE (76.10%) by 3%. ResNet50's deeper residual architecture extracts richer feature representations even with a fully frozen backbone.
- Both pretrained models significantly outperformed the Custom CNN (48.70%), which was trained from scratch for 20 epochs.
- The Custom CNN training log is included below for reference.

Custom CNN Training Log

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============================================================
TASK: Custom CNN Baseline
============================================================
Trainable parameters (Custom CNN): 815018
[CustomCNN] Epoch 01/20 | Train Loss: 2.0612 Acc: 0.2384 | Val Loss: 1.8486 Acc: 0.3110
[CustomCNN] Epoch 02/20 | Train Loss: 1.9434 Acc: 0.2844 | Val Loss: 1.7928 Acc: 0.3430
[CustomCNN] Epoch 03/20 | Train Loss: 1.8790 Acc: 0.3078 | Val Loss: 1.7078 Acc: 0.3940
[CustomCNN] Epoch 04/20 | Train Loss: 1.8282 Acc: 0.3324 | Val Loss: 1.6798 Acc: 0.3810
[CustomCNN] Epoch 05/20 | Train Loss: 1.7940 Acc: 0.3352 | Val Loss: 1.6756 Acc: 0.3920
[CustomCNN] Epoch 06/20 | Train Loss: 1.7748 Acc: 0.3410 | Val Loss: 1.6252 Acc: 0.3950
[CustomCNN] Epoch 07/20 | Train Loss: 1.7353 Acc: 0.3616 | Val Loss: 1.5821 Acc: 0.4040
[CustomCNN] Epoch 08/20 | Train Loss: 1.7109 Acc: 0.3648 | Val Loss: 1.5678 Acc: 0.4250
[CustomCNN] Epoch 09/20 | Train Loss: 1.7068 Acc: 0.3768 | Val Loss: 1.5488 Acc: 0.4320
[CustomCNN] Epoch 10/20 | Train Loss: 1.7085 Acc: 0.3772 | Val Loss: 1.5490 Acc: 0.4390
[CustomCNN] Epoch 11/20 | Train Loss: 1.6668 Acc: 0.3868 | Val Loss: 1.5432 Acc: 0.4260
[CustomCNN] Epoch 12/20 | Train Loss: 1.6649 Acc: 0.3948 | Val Loss: 1.5151 Acc: 0.4350
[CustomCNN] Epoch 13/20 | Train Loss: 1.6379 Acc: 0.3992 | Val Loss: 1.4829 Acc: 0.4480
[CustomCNN] Epoch 14/20 | Train Loss: 1.6328 Acc: 0.3964 | Val Loss: 1.4523 Acc: 0.4570
[CustomCNN] Epoch 15/20 | Train Loss: 1.6201 Acc: 0.4032 | Val Loss: 1.4743 Acc: 0.4470
[CustomCNN] Epoch 16/20 | Train Loss: 1.6330 Acc: 0.4042 | Val Loss: 1.4641 Acc: 0.4510
[CustomCNN] Epoch 17/20 | Train Loss: 1.5903 Acc: 0.4190 | Val Loss: 1.4485 Acc: 0.4600
[CustomCNN] Epoch 18/20 | Train Loss: 1.5914 Acc: 0.4206 | Val Loss: 1.4387 Acc: 0.4820
[CustomCNN] Epoch 19/20 | Train Loss: 1.5934 Acc: 0.4280 | Val Loss: 1.4343 Acc: 0.4610
[CustomCNN] Epoch 20/20 | Train Loss: 1.5844 Acc: 0.4236 | Val Loss: 1.4458 Acc: 0.4870

CustomCNN — Best Val Acc: 0.4870 | Time: 14.7 min
```

The Custom CNN trained from scratch for 20 epochs and reached only 48.70% accuracy, while both frozen pretrained models exceeded 76% within 10 epochs. This demonstrates the significant advantage of transfer learning when training data is limited.

Task 4: Fine-Tuning with VGG16

Setup

- Model: VGG16 pretrained on ImageNet with the last 5 layers unfrozen.
- Trainable parameters: 111,689,738.
- Epochs: up to 20 and early stopping triggered at Epoch 13.

Training Log

```
============================================================
TASK: Fine-Tuning — VGG16 (Last 5 Layers + Improvements)
============================================================
Trainable parameters (VGG16 Fine-Tuning): 111689738
[VGG16-FT] Epoch 01/20 | Train Loss: 1.8919 Acc: 0.4140 | Val Loss: 1.2116 Acc: 0.6080
[VGG16-FT] Epoch 02/20 | Train Loss: 0.9170 Acc: 0.7050 | Val Loss: 0.7770 Acc: 0.7210
[VGG16-FT] Epoch 03/20 | Train Loss: 0.6370 Acc: 0.7856 | Val Loss: 0.6149 Acc: 0.7700
[VGG16-FT] Epoch 04/20 | Train Loss: 0.5004 Acc: 0.8276 | Val Loss: 0.5606 Acc: 0.7920
[VGG16-FT] Epoch 05/20 | Train Loss: 0.4492 Acc: 0.8420 | Val Loss: 0.5078 Acc: 0.8200
[VGG16-FT] Epoch 06/20 | Train Loss: 0.3836 Acc: 0.8672 | Val Loss: 0.4624 Acc: 0.8450
[VGG16-FT] Epoch 07/20 | Train Loss: 0.3432 Acc: 0.8898 | Val Loss: 0.4543 Acc: 0.8470
[VGG16-FT] Epoch 08/20 | Train Loss: 0.2933 Acc: 0.8980 | Val Loss: 0.4754 Acc: 0.8450
[VGG16-FT] Epoch 09/20 | Train Loss: 0.2836 Acc: 0.9038 | Val Loss: 0.4786 Acc: 0.8420
[VGG16-FT] Epoch 10/20 | Train Loss: 0.2416 Acc: 0.9184 | Val Loss: 0.4640 Acc: 0.8440
[VGG16-FT] Epoch 11/20 | Train Loss: 0.2226 Acc: 0.9232 | Val Loss: 0.4717 Acc: 0.8460
[VGG16-FT] Epoch 12/20 | Train Loss: 0.1846 Acc: 0.9378 | Val Loss: 0.4727 Acc: 0.8440
[VGG16-FT] Epoch 13/20 | Train Loss: 0.1688 Acc: 0.9458 | Val Loss: 0.4870 Acc: 0.8410
  ↳ Early stopping triggered at epoch 13

VGG16-FT — Best Val Acc: 0.8470 | Time: 11.5 min
```

Results and Analysis

- Best Validation Accuracy: 84.70% and Test Accuracy: 84.70%
- Training accuracy reached 94.58% by Epoch 13, while validation accuracy plateaued at 84.70% from Epoch 7 onward.
- The growing gap between training and validation accuracy from Epoch 8 onward is indicative of mild overfitting.
- Early stopping triggered at Epoch 13 as validation loss ceased to improve.
- Fine-tuning VGG16 improved accuracy by 8.6% over frozen VGG16 feature extraction.

Although fine-tuning VGG16 produced a notable improvement over its feature extraction counterpart, the large number of trainable parameters (111M) relative to the small training set (5,000 samples) contributed to overfitting. The training and validation accuracy diverged progressively from Epoch 8, with validation accuracy stagnating while training accuracy continued to rise. ResNet50 fine-tuning remained superior due to its skip connections, which provide implicit regularisation during training.

Task 5: Improvements to Fine-Tuning

The following modifications were applied to improve fine-tuning performance, particularly for VGG16, where overfitting was observed:

```
1. Differential Learning Rates
   - Backbone unfrozen layers : lr = 5e-5 / 1e-4
   - New classifier head      : lr = 1e-3

2. Dropout Regularization (Prevent Overfitting)
   - Dropout(0.5) after first FC
   - Dropout(0.4) after second FC

3. Early Stopping (patience=6)
   - Restores best-validation-accuracy weights

4. ReduceLROnPlateau Scheduler
   - factor=0.3 / 0.5, patience=3
   - Reduces LR when val_loss plateaus

5. ImageNet Normalization + Data Augmentation
   - RandomHorizontalFlip, RandomCrop
   - ColorJitter (brightness, contrast, saturation)

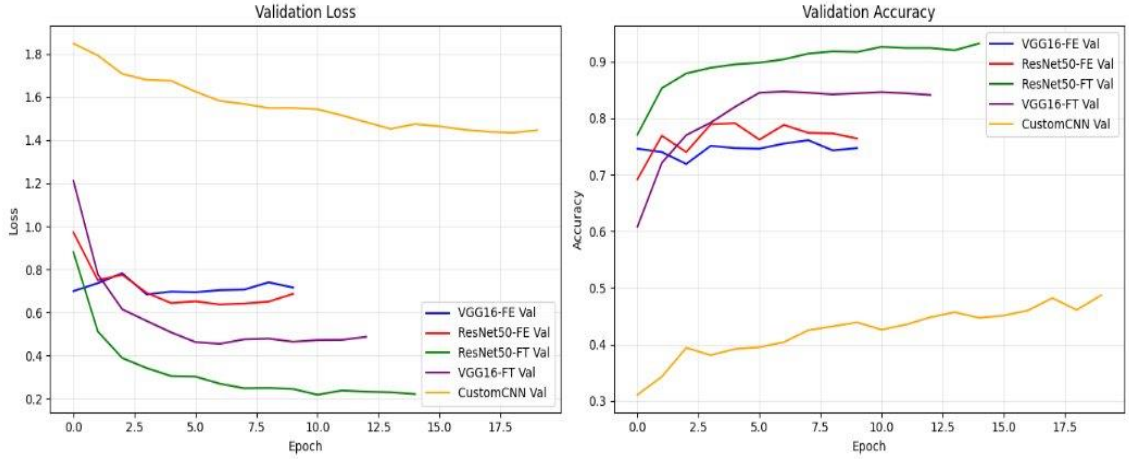
6. More Unfrozen Layers
   - Last 5 feature/conv layers unfrozen
   - Larger part of backbone adapts to CIFAR-10

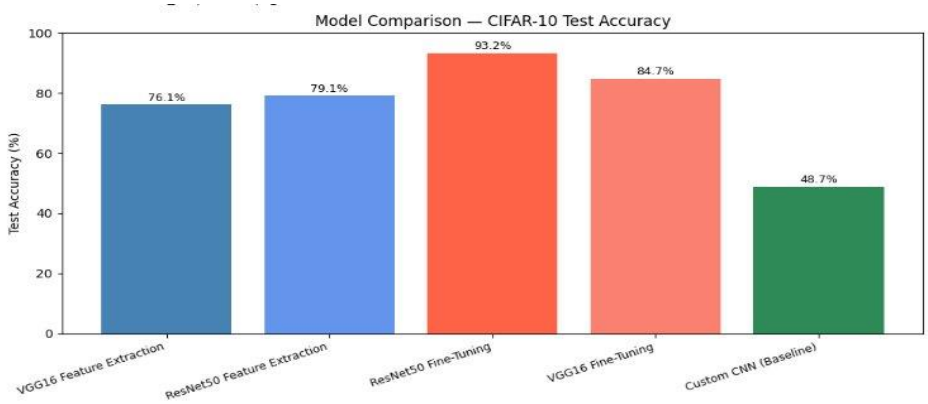
7. SGD + Nesterov Momentum (VGG16-FT)
   - Better generalization than plain Adam
   - weight_decay=1e-4 as L2 regularization
```

Modifications and Rationale

- A smaller learning rate (5e-5) was used for the unfrozen backbone layers, while a higher rate (1e-3) was applied to the classifier head to preserve pretrained features while allowing faster learning in new layers.
- Dropout of 0.5 after the first fully connected layer and 0.4 after the second was used to reduce overfitting and improve generalisation.
- Training was stopped automatically when validation accuracy stopped improving (patience = 6), ensuring the model does not overfit and that the best weights are retained.
- The learning rate was reduced when validation loss plateaued (factor = 0.3–0.5, patience = 3), allowing finer adjustments during later training stages.
- ImageNet normalisation along with random horizontal flipping, random cropping, and color jittering was applied to increase data diversity and reduce overfitting.
- The last five convolutional layers of the backbone were unfrozen to allow better adaptation to the CIFAR-10 dataset. SGD with Nesterov momentum was used instead of Adam, along with L2 regularisation (weight decay = 1e-4), as it generally provides better generalisation during fine-tuning.

Final Results and Comparison





Final Test Accuracy — All Models

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STEP 4 & 5: Final Evaluation & Comparison
============================================================
VGG16-FE — Test Accuracy: 0.7610 (76.10%)
ResNet50-FE — Test Accuracy: 0.7910 (79.10%)
ResNet50-FT — Test Accuracy: 0.9320 (93.20%)
VGG16-FT — Test Accuracy: 0.8470 (84.70%)
CustomCNN — Test Accuracy: 0.4870 (48.70%)
```

Model	Test Accuracy	%
ResNet50 Fine-Tuning	0.9320	93.20%
VGG16 Fine-Tuning	0.8470	84.70%
ResNet50 Feature Extraction	0.7910	79.10%
VGG16 Feature Extraction	0.7610	76.10%
Custom CNN (Baseline)	0.4870	48.70%

Conclusion

Fine-tuning consistently outperformed feature extraction across both architectures, confirming that allowing the pretrained backbone to adapt to the target dataset is beneficial. ResNet50 achieved higher accuracy than VGG16 in both modes, attributed to its residual connections and more compact parameterisation. The Custom

CNN trained from scratch failed to reach competitive accuracy even after 20 epochs, highlighting the importance of transfer learning when training data is limited. The application of differential learning rates, dropout, early stopping, data augmentation, and scheduler-based learning rate decay are effective strategies for improving fine-tuning performance and controlling overfitting.